

Jonathan Zeng

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EDUCATION

Yale University

Expected Graduation Date: May 2028

Bachelor of Science in Computer Science and Mathematics

- **GPA:** 3.93/4.0
- **Relevant Coursework:** Systems Programming and Computer Organization, Data Structures and Programming Techniques, Building Game Engines, Computational Vision, Analysis 1, Discrete Mathematics, Linear Algebra

EXPERIENCE

Student Researcher

June 2025 – Present

Computer Graphics Group, Yale University

- Designing a faster Wavefront .OBJ parser to load and reload 3D objects using multi-threading and SIMD
- Testing the effects of various C++ optimizations on performance, such as thread affinity, profile-guided optimizations, and memory-mapped files, resulting in a 41% performance increase over other parsers
- Communicating results with Professor Mike Shah and other student researchers to write paper for conferences.

Software Engineer

June 2025 – September 2025

Solis Consulting

- Building RoofRoute with Kotlin in Android Studio, an app for roof repair canvassers to find an optimal route of homes to canvass based on parcel and storm data
- Managing backend database on Render.com containing GIS parcel data for over 15 NC counties
- Designing the UI/UX with Jetpack Compose and implementing Google Maps API and voice transcription features

Student Researcher

June 2023 – March 2024

I³T Research Lab, ECE Department, Duke University

- Created and analyzed a game-engine based dataset for monocular Simultaneous Localization and Mapping (SLAM) algorithms using OpenCV for python and the Unreal game engine
- Conducted experiments on various SLAM parameters to find optimal number of features, levels, and scale factor
- Presented poster at NC Science & Engineering Fair, SNCU Research & Creativity Symposium, and lab meetings

Teaching Assistant

August 2023 — May 2024

NCSSM Computer Science Department

- Assisted students in all computer science courses with office hours, events, and online tutorials
- Updated Computer Science Department website and social media with opportunities and newsletters
- Wrote and taught weekly forum classes for 15-20 students on topics like computer vision and generative AI

PROJECTS

Isotetris | D Programming Language

- Designed and developed an original game/game engine featuring an 3D-twist on Tetris and a built-in editor
- Utilized D and SDL to create core engine for isometric rendering logic and custom sound/assets manager.

Shader Lab | JavaScript

- Created website showcasing 3D shapes and shaders using Three.js for WebGL integration
- Wrote custom shaders using GLSL and UI layout for shader options with HTML and CSS

ACTIVITIES & AWARDS

- Amoriem Labs (Yale Game Development Club) Programming Team Lead
- Yale Admissions Student Ambassador (Paid Student Job)
- American Invitational Mathematics Exam (AIME) Qualifier x 4
- High School Mathematical Contest in Modeling (HiMCM) Finalist (top 5%)

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, Kotlin, Lua, D, C#, HTML, CSS

Tools: Git, Linux, Visual Studio, Unity, Unreal, Godot, IntelliJ, Jupyter

Libraries: OpenGL, PyTorch, TensorFlow, SDL, OpenCV, NumPy, Matplotlib, Three.js, Pandas